

Effective Connectivity Predicts Learning Gains in Children: A Dynamic Causal Modeling Study

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Abstract

The ability to detect neural markers of learning-related change is essential for evaluating educational interventions. However, it remains unclear whether regional brain activation or connectivity provides a more sensitive measure of training-effects. In this fMRI study, we reanalyzed data from 54 six-year-olds who underwent one of several cognitive training-programs targeting working memory(WM), mathematics, or both. While conventional GLM analyses revealed no significant longitudinal changes in BOLD-activity, dynamic causal modeling(DCM) of effective connectivity within a fronto-parietal network identified distinct training-related changes. Connectivity patterns differed systematically across training-groups and were significantly associated with individual differences in mathematical improvement. Notably, increased fronto-parietal coupling predicted gains in mathematical performance, while changes in self-inhibition in visual and frontal regions were linked to WM training. These findings suggest that effective connectivity may serve as a more sensitive and functionally meaningful marker of neuroplasticity than traditional activation-based analyses and highlights the potential of modeling network dynamics.

1. Introduction

Understanding how educational interventions influence the developing brain is a central goal of developmental- and educational cognitive neuroscience, with a lot of focus on mathematical abilities (Vogel & De Smedt, 2021). Neuroimaging has the potential to evaluate the neural mechanisms underpinning learning and to identify markers of effective training (Gabrieli et al., 2015). However, it remains unclear whether changes in brain activation or in inter-regional connectivity offer a more sensitive index of training-related plasticity (Constantinidis & Klingberg, 2016; Cortes et al., 2022).

Prior studies have shown that working memory (WM) and math-related interventions can alter brain activity (Olesen et al. 2004, Westerberg 2007, Supekar, K., et al., 2013, Nemmi et al. 2016, Rosenberg-Lee et al., 2018), increase structural or functional connectivity (Takeuchi et al., 2010; Jolles et al., 2013; Supekar, K., et al., 2013, Astle et al., 2015; Thompson et al., 2016, Gabrieli 2016, Lidström et al. 2022), or affect both (Rosenberg-Lee et al., 2018, Cortes et al.,

2022). However, a systematic review found inconsistent patterns of directionality in BOLD activity changes across WM training studies, suggesting that neural adaptations may be nonlinear or heterogeneous (Schiöth, 2020). These inconsistencies highlight the need for models that capture network-level reorganization, such as changes in effective connectivity.

Dynamic causal modeling (DCM) provides a framework for quantifying directed interactions between brain regions (Friston et al., 2003), making it well-suited to investigate training-related changes in network dynamics. In this study, we reanalyzed fMRI data from 54 six-year-old children who underwent different combinations of WM and number-line math training (Nemmi et al., 2016), where participants completed a visuospatial WM task in the scanner before and after training. Our primary aim was to compare task-related BOLD activation and effective connectivity as candidate neural markers of training effects. We further examined whether longitudinal changes in connectivity could predict behavioral gains in mathematics, providing insight into the neural mechanisms of cognitive transfer between WM and mathematics.

2. Materials and Methods

2.1 Participants & Training

This study is a follow-up to the 2016 paper by Nemmi et al., using the same dataset. The participants were all typically developing children of 6 years of age. The original study included 239 children in total, collected in 2014-2015. These children had been training using the Vektor working memory training app ~30min/day for ~8 weeks, and a subsample of them (the focus of this analysis) performed an fMRI WM task before and after the training.

The Vektor training consisted of two parts, during the first week, which was identical for all participants, baseline mathematics and working memory performance was established. Starting in week 2 each child was assigned into one of four training conditions. Each condition had two components, and training time was split 50/50 between them. The groups with their training contents were: 1. Working memory training and reading (WMT)(n=15), 2. number-line training and reading (NLT)(n=12), 3. working memory and number-line training (WMT&NLT)(n=13), 4. reading and reading (controls)(n=14). (Going forward, these three non-control training groups will be referred to as 'WMT', 'NLT' and 'WMT&NLT').

Vektor is a cognitive training application that is free by invitation, developed by Torkel Klingberg and designed for 6-year old children with no prior math skills required. The app dynamically scales the difficulty of tasks according to the skill level of the children by trying to achieve and maintain a 60% rate of correct responses for WM and 90% for mathematics. The WMT module includes four different tasks of VSWM. All include a sequence of positions that must be maintained in memory, and later indicated by touching the screen. Difficulty levels correspond to the number of positions in the sequence to be remembered.

The NLT module includes tasks where you drag your finger along a numberline to indicate the answer. It begins with simply matching a number presented on the screen, then moving on to negative numbers, addition, and subtraction. Additional difficulty was included by varying the length of the number line, and by reducing the number of ticks on the numberline. Around 15% of the total math time was also dedicated to a ten-pals task (Butterworth et al., 2011). In ten-pals they were presented with a bar of blocks of length 0-10, and the task was

then to select a complementary bar from the right of the screen such that the total length of the two equals 10.

The reading control task was adapted from the GraphoGame (Lyytinen et al., 2009, Brem et al., 2010). The task is to associate letters and letter combinations with sounds, followed by short whole words, and finally, easy crossword-puzzles.

56 participants had functional neuroimaging data. In this analysis, we excluded two subjects, one with only 2 training days, (insufficient for establishing math improvement rate), and one affected by very severe motion, and where the top of the brain was outside the field of view.

Imaging experiments using young children are inherently difficult, and more head motion is to be expected. To assess the impact of motion artifacts, we analyzed task activation in SPM12 with and without ARTrepair motion correction. The resulting activation patterns remained consistent, confirming the robustness of our results. Voxel-wise correlation between contrast image results with/without ARTrepair toolbox(0.5 threshold) in the ROIs(see 2.5)(5mm spheres) was $r = 0.956$. Means of the contrast images were 1.009 vs. 1.306(arbitrary units) with a voxel-wise mean difference of 0.297 ± 0.123 (SD). We calculated the Tukey mean-difference 95% limits of agreement(LoA) between the two images, which tells you the interval where 95% of the differences are expected to lie, assuming the differences are normally distributed. The LoA was ± 0.24 , with 15 voxels outside this range, equating to 3.7%. Taken together, this indicates that using ArtRepair would change the estimated levels of activity, but not so much the pattern, meaning that the results would differ systematically, but consistently, across voxels). That we found plausible GLM results matching previous findings (see 3.2) and were able to match them with relevant behavioral correlates also suggest validity.

We calculated each participant's mathematical improvement rate by analyzing their number-line task performance over time. We plotted the average difficulty level of correctly solved trials per day and computed a linear approximation between the first session and the highest performance peak to avoid plateau effects.

2.2 fMRI Task

The participants performed a load 2 or load 4 VSWM task where they had to remember the positions of either 2 or 4 objects (planet earth pictures) appearing in sequence (500ms presentation, 500ms between presentations) within a 4x4 grid , and after a further delay of 500ms, respond yes or no to indicate whether a final position, cued by a question mark, was in any of the spaces previously occupied by one of the symbols in the memory sequence (Söderqvist et al., 2010, Dumontheil & Klingberg, 2012). Two sessions of this task were performed with a break in between at each timepoint (before and after training) for a total of 4 sessions of data. There were also load 2 and load 4 control tasks where images of the sun always appeared in the corners (upper corners for load 2, and all corners for load 4) and the instruction was to always respond with a NO to a centrally placed question mark. Each participant did two runs of 32 trials (8 for each condition) separated by 2 second inter-trial intervals.

2.3 Imaging Parameters

GE Discovery 3T MR750 scanner using 8-channel phased array receiving coil. T1-weighted anatomical image parameters: 1-mm³ isotropic voxels. TE = 3.06ms, TR = 7.9ms, TI = 450ms, FoV = 24cm, 176 axial slices, flip angle = 12°. EPI image parameters: 3mm³ isotropic voxels. TE = 30ms, TR = 2200ms, FoV = 22cm, 46 axial slices, flip angle = 70°, 130 volumes total.

2.4 SPM GLM Analysis

A GLM analysis was performed using SPM12 to test whole-brain measures of longitudinal change, and as the basis for our DCM analysis. Each GLM contains up to 4 sessions (depending on completion), each containing columns encoding onsets and durations of correct trials only for the load 2 and load 4-tasks, the load 2 and 4 control tasks, 6 columns of SPM realignment parameters to control for motion, and one constant column per session to capture session effects. Results from this GLM analysis are presented thresholded at $p < 0.001$ uncorrected for display purposes (see table 1, figure 2 and appendix 2). The group analysis of the WM task vs control task was corrected for the number of volumes for each participant by means of a nuisance regressor.

2.5 ROI Selection

Regions of interest (ROIs) were selected from the GLM analysis by contrasting the two WM tasks with the two control tasks and selecting the 3 most significant regions, and their corresponding left hemisphere activations in the case of superior frontal gyrus (SFG) and superior parietal lobule (SPL), resulting in 5 total ROIs. Data for the DCMs were extracted from these ROIs for each subject separately by centering a sphere of radius 10mm around the GLM coordinate and allowing a smaller 5mm radius sphere to move to the local maximum within it. The mean of the first eigenvalues of the voxels within the small sphere were then calculated for each timepoint, producing a time-series for each ROI per participant. (The peak GLM coordinates of the ROIs are given in table 1). The group level beta values from the 5 ROIs were also extracted to look for correlations between mean activity levels with math improvement rate (mean first eigenvalues, 5mm sphere).

2.6 DCM Setup

The structure (A-matrix) of our DCM was to connect bidirectionally between the left and right SPL nodes and the left and right SFG nodes. Bidirectional connections were then added between the SPL and SFG within each hemisphere, and finally, the node in the left visual cortex was connected bidirectionally to both SPL nodes. Each subject had a DCM fitted to their individual ROI data by concatenating all four MRI sessions of WM task onsets and durations using the SPM concatenation function. This treats the data as one long experiment, with columns accounting for session effects. We then added a column coding for time-point (0s for the first two sessions scanned before training and 1s for the two sessions scanned after training). This approach allows us to later estimate changes in connectivity as effects of time and training conditions at the group level using covariates. The driving experimental input (C) was entered into the model via the visual cortex node, and all connections within the model

were allowed to be modified by the time-encoding condition column (B-matrix parameters). Quality of fit was evaluated by looking at % variance explained in the ROI time-series by the DCM for each subject.

2.7 GCM (PEB)

Group connectivity modeling (GCM) in the DCM paradigm is performed using a model called Parametric Empirical Bayes (PEB). This is analogous to a 2nd level analysis in a GLM, but we enter individual DCMs instead of SPMs. One of the advantages of the PEB model is that the individual DCMs are weighted by their confidence, meaning that worse fitting individual models will contribute less to the group connectivity estimates. The design of the models at this stage is very straight forward, consisting only of a constant and one column with a covariate encoding what we are trying to correlate with our group connectivity parameters. We tested 3 covariates, each coding for which participants belonged to the WMT, the NLT or the WMT&NLT group, respectively.

Since we have a covariate in our PEB models, the resulting parameters are presented in two sets: the first represents the average connectivity, excluding that associated with the covariate, and the second, representing connectivity attributed to the covariate (a particular training condition in our case). Each set has two types of parameters: A parameters, encoding the “commonalities” i.e. what is common across the whole set, and B parameters, encoding changes in connectivity brought on by external experimental manipulation, (the time-point column in the DCMs in our case, thereby encoding changes in connectivity after training). This combination of our PEB training covariate with the B parameters encoding longitudinal change enables us to estimate what connectivity changes over time are correlated with each of the three training types.

We also used Random Effects (RFX) Bayesian Model Comparison (BMC) to investigate which covariate provided the most explanatory power to the PEB group model of connectivity. BMC looks at the distribution of the best model to explain the data in a group of subjects (“how often is model k the best?”) and RFX analysis allows what is the best model to vary between subjects and simply counts what the most common model is. We report the Bayes Omnibus Risk (BOR) and Protected Exceedance Risk (pxp). The BOR is a normalized Bayes factor, the probability of wrongly picking H1 over H0, or in other words, the probability of the observed preference in your sample for model k occurring by chance. The BOR can be compared to any error rate, (e.g. 5%), but tells us nothing about which model is “best”. It is an omnibus test, and a measure of risk when later looking at individual exceedance probabilities for models. The ppx is a Bayesian model average of the exceedance probability (xp) over H1 and H0, as modified by the BOR. It quantifies the probability that a given model is more frequent than the others, above and beyond chance, where xp is the probability that the k:th model is more common (frequent) than the others (given that H1 is true, i.e. true model rates are not equal)(Rigoux et al. 2014).

2.8 Parameters of Interest & LOO Analysis

For each PEB model, we selected the strongest parameters by a stepwise lowering of the probabilistic threshold in 5% decrements, starting with a probability $P_p > .99$ of the parameter being non-zero, and then choosing all parameters significant at that threshold or above. To quantify the predictive validity of our selected DCM parameters, we then performed post-hoc

statistical analysis of the significance of the training covariates, as evaluated by Leave-One-Out analysis (LOO) using a GLM over our selected parameters as predictor, as per DCM praxis.

After looking at our three training conditions in this way we also wanted to predict the mathematical improvement rates of the participants using only the B parameters indicated in our three PEB models. To do this we first had to use a subsample of participants who all had fMRI data at both timepoints and NLT(n=12) or WMT&NLT(n=13), resulting in a total sample of n=25. We then implemented a scheme akin to backwards stepwise regression, where we started with a LOO analysis where all our selected B parameters were used in the predictive model. We then calculated LOO models where one of the parameters was excluded in turn, and in each generation we discarded the worst performing parameter in terms of adding predictive validity. This produced 5 generations of models. (see appendix A for the complete list of models tested and the correlations between predicted and actual values of mathematical improvement rate.)

3. Results

3.1 Behavioral Data

First, we analyzed behavioral data from the larger study by Nemmi for all participants with NLT in their training plan (n=155) in order to look for differences in improvement rate between the WMT&NLT group and the NLT group. We used the daily training data from the Vektor app to calculate the rate of mathematical improvement by computing the average level of correctly answered NLT trials each day for all participants, and then plotting the average of the resulting time-series for the respective training groups (figure 1). Fitting a linear regression to the two groups, we found that the WMT&NLT group showed a significantly greater rate of improvement (slope) compared to the NLT&reading group (1.107 vs 0.948 p(153)=0.0415).

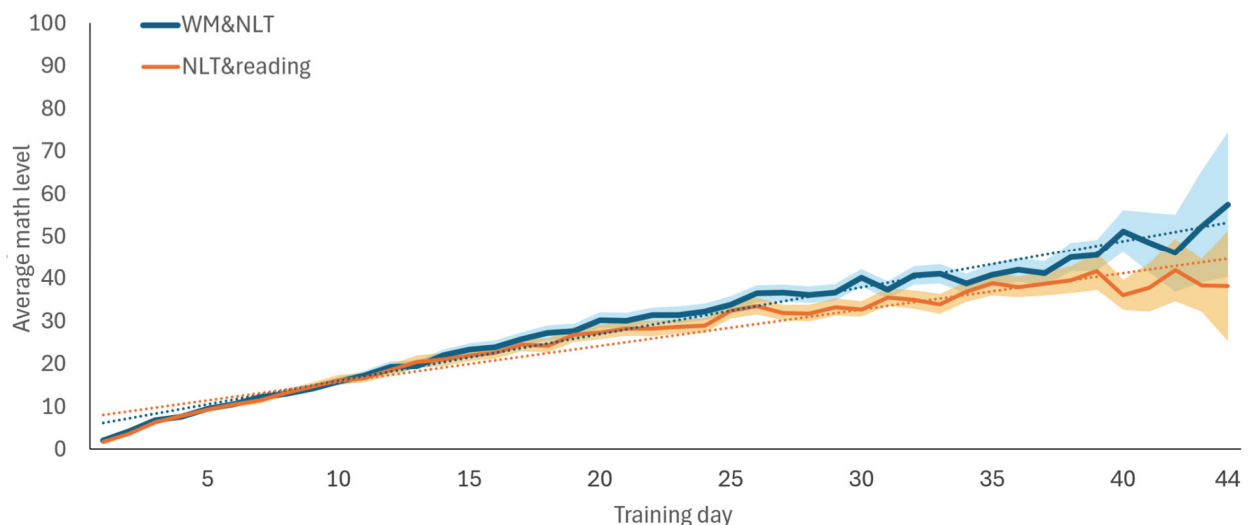


Fig. 1. Average difficulty level of correct NLT trials at each day of training for the combined WMT&NLT group (blue) vs. the NLT group (orange) in the full dataset from Nemmi 2016. Dashed lines=Linear regression slopes. Shaded areas=Standard Error of the Mean.

3.2 Analysis of Changes in WM-related BOLD Magnitude

Our re-analysis of the task fMRI data in our whole sample of $n=54$, combining both the pre- and post-training fMRI data, revealed two significantly activated peaks at the whole-brain Family-Wise Error-corrected (FWE) threshold of $p<0.05$ when contrasting the WM task with the control task: Left Calcarine cortex $[-8 -90 -4]$ $T=7.31$, $p(\text{FWE})<0.000$, $k=187$ voxels, and Right Superior Frontal Gyrus $[28 6 64]$ $T=5.76$, $p(\text{FWE})=0.035$, $k=1$ voxel. (See Table 1 & figure 2 for uncorrected clusters). Post- minus pre-levels of BOLD activity in the main effect of task (WM minus control) was analyzed for each of the 5 ROIs. No significant correlation (Two-tailed) was found between BOLD activity changes in the ROIs and math improvement rates (lCalc $r=0.107$ $p=0.431$, rSFG $r=-0.058$ $p=0.672$, rSPL $r=0.023$ $p=0.863$, ISPL $r=-0.007$ $p=0.959$, ISFG $r=-0.156$ $p=0.251$).

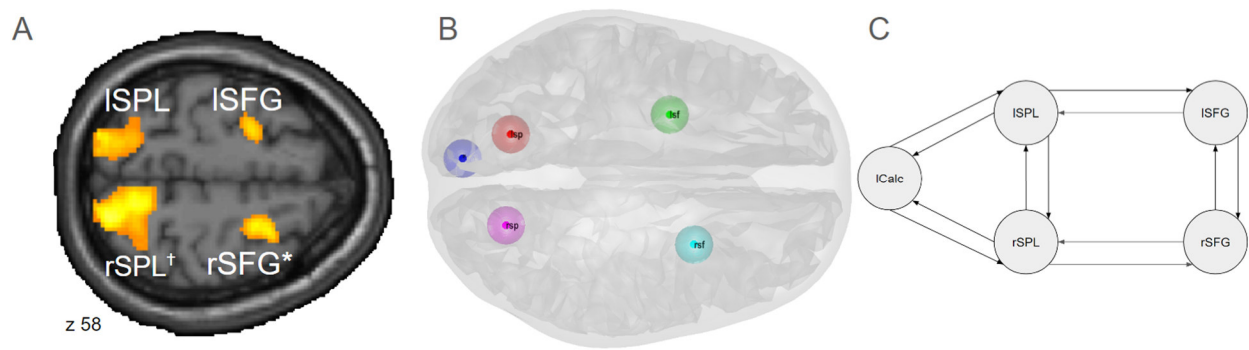


Fig. 2. A=Axial slice through four of the clusters used in the DCM, GLM results presented at $p(\text{uncorrected})<0.001$. $*$ = $p(\text{FWE})<0.05$. $†$ = $p(\text{FEW})<0.06$. B=location of DCM ROI peak coordinates on glass brain. C=Schematic of our DCM showing connections between the nodes.

Region	n voxels	p(FWE-corr)	T	xyz (mm)
l Calcarine cortex (lCalc)	1989(187)	<0.000*	7.317	[-8 -90 -4]
r Superior frontal gyrus (rSFG)	499(1)	0.035*	5.765	[28 6 64]
r Superior parietal lobule (rSPL)	2356	0.055†	5.621	[20 -72 58]
r SFG medial segment	995	0.057†	5.614	[8 28 40]
r Frontal operculum	477	0.059†	5.599	[32 22 10]
l Cerebellum Exterior	565	0.2	5.178	[-14 -60 -48]
l Superior parietal lobule (lSPL)	709	0.241	5.108	[-18 -70 62]
l Cerebellum Exterior	79	0.345	4.963	[-28 -68 -32]
r Precentral gyrus	629	0.456	4.838	[54 12 30]
l Superior frontal gyrus (lSFG)	284	0.623	4.673	[-26 -4 50]

Table 1. Selected GLM Peak Statistics

Top results in order of significance of main effect of WM minus control for the whole sample (pre- and post-training fMRI data combined), showing peak coordinates (excluding sub-peaks) and extent of activations (n voxels) at $p < 0.001$ uncorrected (and $p(\text{FWE}) < 0.05$ in parentheses for clusters surviving FWE correction). Peak statistics reported at $p(\text{FWE}) < 0.05$. Regions in bold indicate inclusion in the DCM. * = significant at $p(\text{FWE}) < 0.05$. † = $p(\text{FEW}) < 0.06$. For the full GLM table, see appendix B.

Next, looking for differences in brain activity during the WM task between participants with and without WM training in Vektor (WM $n=28$, no WM $n=26$), no significant differences were found (Strongest peak $T=5.31$, $p(\text{FWE})=0.151$, right cerebellum exterior [10 -76 -26]). We examined training condition differences over time using a 2x4 full factorial design (2 dimensions

of time and the 4 training conditions). No significant differences were observed across training types. We also found no significant interactions between any of the training types and time.

3.3 Training-Related Changes in Effective Connectivity

3.3.1 DCM Quality of fit and Bayesian Model Selection

To evaluate the evidence for our hypothesized DCM structure (figure 2C), we first used BMC to compare it to a 'full' model with all possible connections turned on. This analysis favored the reduced model (pxp = 0.681 vs 0.318, adjusted from xp = 0.95 vs 0.049 by the BOR of 0.597). We therefore chose to continue our analysis using the reduced model. The overall quality of model fit for the reduced model was then evaluated further by analyzing the percent of variance explained for each participant's DCM, and the values were found to be well over the recommended 10% minimum: $44.14 \pm 2.48\%$ (min=39.74, max=53.72), indicating that the reduced DCM structure has good within-subject explanatory power.

Next, we evaluated what covariate structure at the group level best explained our data. Three PEB models incorporating different covariates coding for training content were estimated: WMT, NLT and combined WMT&NLT. These models were compared at the second level, again using BMC. The analysis favored the model with combined WM&NLT (pxp=0.467) over WMT (pxp=0.266) and NLT (pxp=0.268) with BOR=0.508. The high BOR and the very small differences between the runners up indicates that all the models were very similar in explanatory power, but nonetheless the analysis indicates that the combined training explains more of the variance between participants. Accordingly, we chose to proceed by first examining what connectivity correlated with WMT and NLT separately, then following this, looking for effects of connectivity in the combined training group compared to the rest of the participants.

3.3.2 Longitudinal Changes in Connectivity

As expected, the estimated parameters of both the WMT and NLT PEB models showed a robust response of activation on average, both within and between all nodes in the network, during the WM task (A parameters). For the NLT model, the common changes in task response after training (B parameters encoding average change from pre to post excluding variance attributed to the covariate for training-type) were similar to the parameter changes further attributed to the training covariate, as demonstrated by a significant correlation between these two B parameter matrices ($R=0.6709$, $p=0.0032$). The most robust B parameter change due to training ($p>.95$ of being non-zero) was ISFG->ISPL (0.149, $Pp=0.98$), we therefore selected this parameter for use in LOO validation of the NLT training covariate, and the resulting estimates based on this parameter for each subject significantly correlated with the actual covariate scores ($r=0.36$, $p=0.0036$), indicating that our DCM can predict (post-hoc) if a participant had NLT vs. no NLT significantly better than chance from this parameter (predictive validity). For the WMT PEB, there was no significant correlation between the B parameters for the WMT- and the NLT-models: $R=0.06$, $p=0.8135$. The most robust parameter changes after training in the WMT PEB ($Pp>.85$ of being non-zero) were the three inhibitory self-connections: ICalc (-0.095, $Pp=0.86$), ISFG (0.105, $Pp=0.88$) and rSFG (0.095, $Pp=0.86$). LOO analysis using these three parameters were also strong enough to successfully predict WMT vs. no WMT significantly better than chance ($r=0.24$, $p=0.0427$).

Returning to the question of the effects of combined WMT&NLT, recall that the PEB with WMT&NLT as covariate explained more of the data than WMT or NLT alone. We therefore looked at a PEB model with combined training as covariate to look for parameters that covaried with this group. The strongest effect of combined training were the two parameters connecting the rSPL and rSFG nodes (rSPL->rSFG 0.065 p=0.79, rSFG->rSPL 0.087 p=0.83). While the A- and B-parameters together successfully predicted combined training ($r=0.33$, $p=0.007$), the B-parameters alone did not yield significant predictive power.

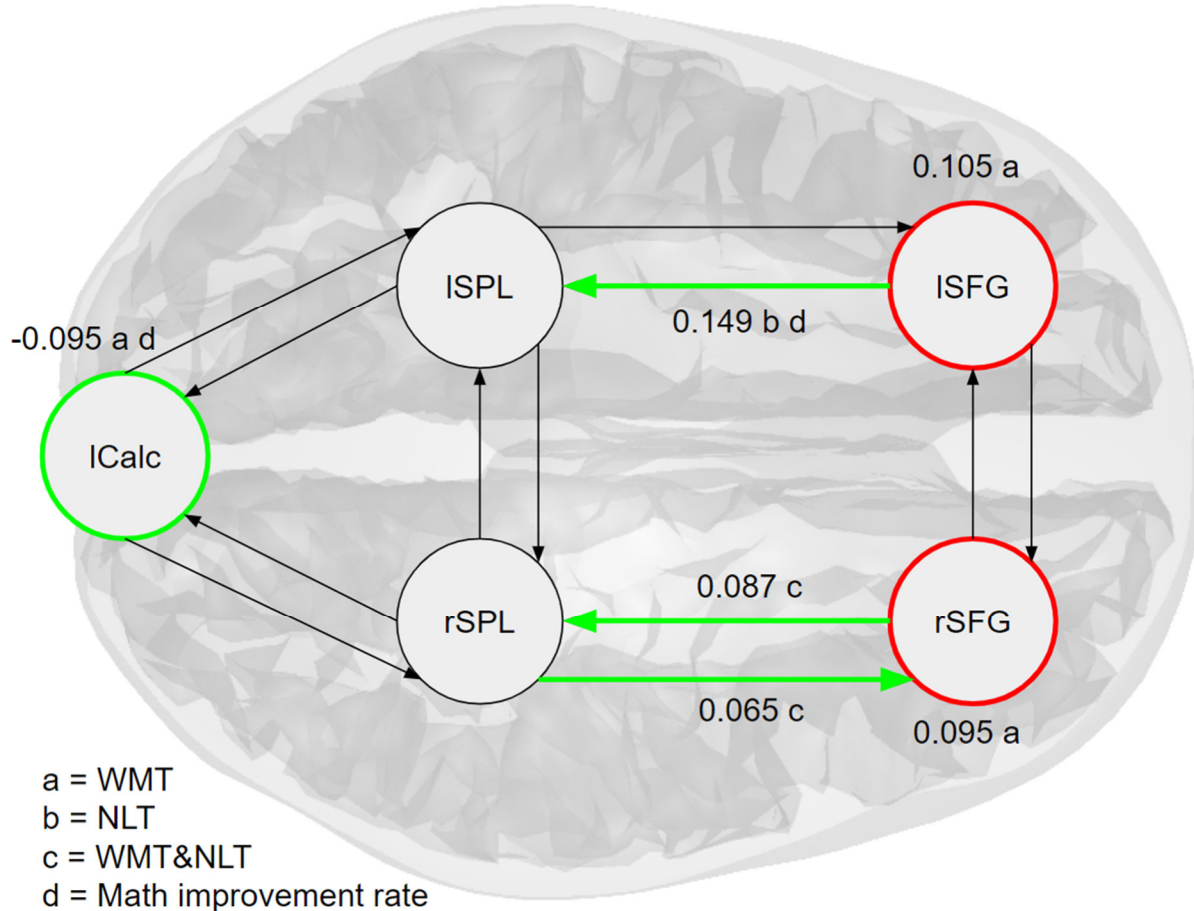


Fig. 3. Significant DCM B-parameters (changes after training). Values=PEB parameter strengths(Hertz). Letters=covariates predicted by LOO analysis. Green=Excitatory. Red=Inhibitory(See discussion for more detailed interpretation).

3.3.3 Relating Connectivity Changes to Behavior

We have shown that these connectivity parameters differ significantly with training conditions, but in order to anchor these findings to behavior we wanted to test if they were strong enough to predict the improvement rate in mathematics during training. Additionally, what our hypothesis predicts is that the changes over time (B-parameters) should capture the significant changes due to training conditions, and therefore we would like to see significant predictions of variables outside the model using B parameters exclusively. To test this, we initially entered all six of the B parameters implicated in training effects into a LOO analysis (ISFG, rSFG, rSPL->rSFG, cuneus and ISFG->ISPL (figure 3)). This model was close to, but

not significant ($r=0.31$, $p=0.068$). Using the systematic stepwise regression approach described in the methods section, we eventually ended up with the best model for predicting mathematical improvement speed during training using the training-associated changes in cuneus and ISFG->ISPL ($r=0.38$, $p=0.031$). This model (and others from earlier generations of the process, see appendix A) show that these changes in connectivity strength estimates can predict behavioral data outside of the group connectivity models, i.e. a behavioral variable not used in either estimating the parameters in the PEB or in selecting what connections to include in the LOO validation. Using Fisher's Z-scores, we directly compared the correlation strength between our best predictive DCM parameters and math improvement rate to the correlations between changes in activation levels during the task (as described in Section 3.2) and math improvement rate. The results showed that the DCM-derived prediction was significantly stronger than activation level prediction in two of the nodes, with a third node trending towards significance (DCM $r >$ cuneus: $p = 0.126$, rSFG: $p = 0.036^*$, rSPL: $p = 0.07$, ISPL: $p = 0.055^\dagger$, ISFG: $p = 0.014^*$).

4. Discussion

4.1. Aims & Findings

The aim of this study was to investigate and compare brain correlates of cognitive and mathematical intervention in children. More specifically, we asked if BOLD measurements of task related brain activation or connectivity (using DCM) were better suited to evaluate neural change and transfer effects.

Behaviorally, we found that a combined WM and Math training resulted in a faster rate of math improvement than training Math and reading, consistent with prior findings of transfer from VSWM training to learning of mathematics (Nemmi et al. 2016, Zhang et al. 2018, Swanson & McMurran 2018, Giofrè et al. 2018, Judd & Klingberg 2021, Berger et al. 2025).

We found significant task-related BOLD-activation in the left calcarine cortex and right superior frontal gyrus (whole-brain corrected). However, no significant changes were observed when comparing before and after training. Correlations between task-related activity and math improvement rate were non-significant in all our ROIs, and formal statistical tests indicated that our DCM-derived correlation with math improvement was significantly stronger than two of the activity-based predictors.

Analysis of DCM connectivity revealed a bilateral, fronto-parietal network. In contrast to the task related activity, we found a significant change in connectivity associated with training. Moreover, our connectivity model could predict both training conditions and math improvement rate significantly better than chance. These findings indicate that effective connectivity may serve as a more sensitive marker than task-related BOLD amplitude in assessing neural correlates of cognitive and mathematical training interventions in children.

WM training was associated with decreased local self-inhibition in the calcarine cortex and increased local self-inhibition in the left and right SFG. Math training was associated with increased effective connectivity from L SFG to L SPL during the WM task, and the combined WM and Math training group showed increased connectivity in both directions between rSFG and rSPL. The overall pattern after training therefore seems to indicate an emphasis on

connectivity between the SFG and SPL, with local modulating effects acting on the SFG and a sensitization of the left calcarine cortex.

4.2. Previous Intervention-Studies & Connectivity

Several studies have explored how cognitive interventions targeting working memory, mathematics, or educational attainment can lead to changes in brain connectivity. Research on WM training has shown that it can strengthen the structural connectivity in the fronto-parietal network (Jolles et al. 2013, Takeuchi et al. 2013, Astle et al. 2015, Thompson et al. 2016). One example, using DTI, found that the amount of working memory training correlated with increased fractional anisotropy in white matter regions adjacent to the intraparietal sulcus and the anterior part of the corpus callosum, areas associated with working memory processes (Takeuchi 2010). Combining transcranial direct current or magnetic stimulation (tDCS or TMS) with cognitive training has also been shown to enhance functional connectivity and working memory performance in older adults and suggest the potential for addressing age-related cognitive decline (Kundu et al. 2013, Nissim 2019).

Since mathematics has been established as a close correlate to working memory performance (Vogel & De Smedt, 2021), looking at mathematics learning in the context of connectivity is informative. The absence of mathematical education during adolescence has been linked to reduced brain inhibition levels and frontoparietal functional connectivity in regions critical for reasoning and cognitive learning. This finding underscores the importance of continuous mathematical education for maintaining healthy brain development and cognitive function (Zacharopoulos 2021). Taken together, these findings show that targeted cognitive interventions can induce measurable changes in brain connectivity, which are associated with improvements in specific cognitive functions related to working memory and math performance.

4.3. BOLD Activation vs. Connectivity

Some studies have directly compared task-related BOLD with connectivity changes: Mennes et al. (2010) examined the relationship between resting-state functional connectivity and task-induced BOLD activity during an Eriksen Flanker task, finding that inter-individual differences in resting-state connectivity could predict task-induced BOLD responses. Additionally, Tsvetanov et al. (2018) investigated how both activity and connectivity contribute to inhibitory control across the lifespan. They found that both factors make distinct, complementary contributions to performance, with connectivity's association with inhibitory control differing with age. Studies like these show that both task-based BOLD activity and connectivity are important for understanding brain function, and their relationship can vary among individuals and across different contexts. This suggests that BOLD and connectivity analyses can either complement each other, or one could be preferable over the other depending on studied population and task.

Connectivity changes may serve as a more robust indicator of intervention effects, reflecting dynamic neural interactions rather than isolated regional activity shifts. While BOLD activity reflects regional activation in response to a task, connectivity measures how different regions communicate, which may be more directly linked to the cognitive and behavioral changes we are looking for. Interventions such as pharmacological treatments, cognitive training, or neuromodulation also often aim to modify network-level interactions rather than isolated brain activity (van Balkom 2020). Connectivity may also be more stable over time and

less susceptible to momentary fluctuations (depending on the time scale we are looking at), making it a potentially more reliable marker of intervention-related neural changes, as shown for example by the longitudinal study of working memory development by (Ullman, 2014) that found that cortical activity predicted current capacity while white matter tracts predicted future behavior.

4.4. Previous Connectivity Studies & Transfer-Effects

Both the task related brain activity and connectivity patterns found in this study are consistent with previously published papers investigating VSWM which also highlight the importance of the visual cortex, SPL and SFG. (Olesen 2004, Westerberg & Klingberg 2007, Darki & Klingberg 2015, Gabrieli et al. 2015, Caeyenberghs et al. 2016, Thompson 2016, Constantinidis & Klingberg 2016). Similar patterns of activation and connectivity have also been reported in studies on mathematics (Prescott et al. 2010, Wang et al. 2022) and programming (Floyd et al. 2017, Lidström 2022).

Studies of transfer include Cortes et al. 2022, where they found that students trained on geospatial tasks showed greater longitudinal increases in activation in the anterior intraparietal sulcus and increased connectivity of areas implicated in spatial processing (based on a Neurosynth.org meta-analysis) to a DLPFC region implicated in syllogistic deductive verbal reasoning. They also found that students in the geospatial training group showed greater improvement in reasoning than controls, consistent with far transfer from spatial to deductive reasoning. Another study also showed that working memory training in adults improved other non-trained WM tasks and transferred to tasks of reasoning and inhibition (Caeyenberghs et al. 2016). One of the largest studies of transfer effects to date is Berger et al. 2025. They found that WM training improved not only their WM capacity, but also transferred to geometry, reading, IQ, and self-regulation (executive control). They also highlight the importance of early targeting (the 6-7 age range), and that the effects of the intervention manifests fully up to about 3 years after the intervention, so post-evaluations should ideally be more longitudinal.

When it comes to the composition of the training interventions and transfer, Judd & Klingberg 2021, continuing to iterate on the training used in earlier studies, again found that the ratio of different tasks is significant, and that a mix of VSWM and non-verbal reasoning was the most effective in promoting transfer to mathematical learning. A comprehensive meta-analysis by Sala and Gobet (2019) examined the efficacy of cognitive training programs in producing both near and far transfer effects. They found that while some training programs led to improvements in tasks similar to those trained (near transfer), evidence for far transfer (improvements in distantly related tasks) was limited. However, they define far transfer from cognitive training as enhancements in tasks like reasoning, processing speed, and language and they do not explicitly address the transfer effects of working memory training on mathematics performance. Therefore, it is unclear whether the authors consider the transfer from working memory training to mathematics as far transfer or not. Nevertheless, this suggests that training in fundamental cognitive skills does not consistently result in broad cognitive enhancements across different domains. Taken together, these findings highlight the challenges in designing cognitive training interventions that effectively generalize to untrained tasks,

emphasizing the need for targeted approaches to achieve specific desired transfer effects, and efficient evaluation of them.

4.5. Neurobiological Implications

Although brain imaging of pre-post effects cannot directly discern what neural mechanisms are responsible for the changes in overall activity or connectivity, several candidate explanations have been explored. Changes in brain activity as measured by BOLD GLM statistics could reflect more or less efficient processing in terms of the amount of local network neurons recruited and/or increased or decreased firing rate, but this is also affected by time on-task in the time-window of the GLM, potentially making it somewhat difficult to interpret directly in terms of underlying biology. Conversely, our DCM analysis shows that the plasticity induced by WM and math training primarily emerges as changes in the network dynamics within the frontoparietal networks in both hemispheres, and that task BOLD amplitude might not be sufficient to detect any effects of this training, nor significant differences between various interventions with subtle differences in effect.

Changes in brain connectivity could reflect short- or long-term potentiation of connections. These adaptations are driven by processes like changes in dendritic spine density and myelination (Takeuchi et al. 2010, Gibson et al. 2014, Xin & Chan 2020,), the latter of which is believed to be the major contributing factor when measuring fractional anisotropy using DTI. For instance, studies of myelination in mice show changes after 7-14 days of fear learning (Xin & Chan 2020). They argue that myelination can fine-tune information transmission and timing to various targets in the brain and showed that enhanced performance after training is correlated with fractional anisotropy. Disrupted oligogenesis (decreased birth of new myelinating cells) in the mice were also linked to reduced spatial memory, possibly due to reduced capacity for consolidation or maintenance of learned rules and patterns.

Using MEG, Ericson et al. (2024) has shown that synchronization in the alpha band increases linearly over the period of WM training in the dorsal attention network (primarily in the intraparietal and superior parietal cortices and in the superior frontal gyrus) and that this was associated with transfer effects. This strengthening of synchronization as a marker of both training and transfer in the dorsal attention network aligns with the idea that the functional connectivity changes that we see in this study come as a result of neuronal tuning mechanisms, and is the primary driver of the observed behavioural changes after training. More congruent activity within this frontoparietal network might be the primary goal of this pre-post training network tuning. With regards to time frame, significant changes in effective connectivity 1-2 days after a learning session of only 3 hours has been observed in humans learning computer programming, indicating that measurable task-relevant effective connectivity changes can also develop rapidly in humans with long-term effects (Lidström 2022).

4.6. Interpretation of DCM parameters

We have chosen to present both extrinsic (between-node) and intrinsic (inhibitory self-connectivity) DCM parameters. Extrinsic parameters are relatively straightforward to interpret: positive values indicate excitatory interaction where activity in the originating node causes the activity in the receiving node to increase, and the opposite effect for negative values.

The intrinsic connectivity parameters, on the other hand, are harder to interpret. First, their effect is inverted, where a higher parameter represents more inhibition. Positive self-inhibitory parameters are not inherently "decreasing activity" though, but represent a stabilizing mechanism that keeps activity in check (without these parameters the activity would increase indefinitely over time in the model). A biological interpretation of the intrinsic parameters is that positive self-inhibition reflects the natural decay of neuronal activity within a region due to local inhibitory processes, such as GABAergic mechanisms, and changes in this self-inhibition may be interpreted as more or less local excitability in the region. This tunes the node to how prone it is to persistent activity (Friston et al. 2003, Zeidman et al. 2019).

4.7. Limitations.

Due to the original study's design, each of the four learning conditions consisted of approximately 12 subjects. The fact that the DCMs were calculated individually, and we then used the whole sample to look for average differences in the parameters due to training addresses this limitation since it allowed us to identify interesting connections with more confidence and then evaluated them using our covariates. Similarly, for subjects lacking some of the scanning sessions, they can still contribute to the common connectivity pattern in the PEB model.

The statistical tests of covariates from connectivity parameters in our LOO modeling are post-hoc, thus technically not a prediction of future behaviour, but rather a statistical test of correlations with estimated connectivity changes over time. However, the significance of our post-hoc test results indicate that we should expect to be able to predict future behavior or training conditions from these parameters in an independently collected sample, but this is of course not a guarantee, rather something that should ideally be tested.

Another limitation of this study is that we did not optimize the basic connectivity structure of the DCMs beyond comparing them to a full model, we did however see that individual model evidence was good. We also focused primarily on the B parameters (like in most DCM studies) since they encode the experimental modulations (change over time in our model). If the aim is to refine optimal predictive models however, it is possible that by also incorporating the common connectivity strengths of relevant connections (the A parameters) into the LOO models, even better predictions of future outcomes or conditions would be possible.

4.8. Wider Implications for Educational Neuroscience.

What role, if any, neuroscience has to play in the field of education has been intensely debated (Bruer 1997, Ansari 2006, Matta 2021). The critique mainly falls into one of three categories: 1. That neuroscientific experiments lack ecological validity (i.e. that they will miss crucial factors, or that the results will be highly context-dependent to the experimental conditions only and not predictive of real-world outcomes). 2. That traditional behavioral measures or classroom evaluations better capture the current and future performance of students and therefore questions the relevance of individual research findings. 3. That neuroscientific findings lack complete explanatory models for how their findings might translate into significant effects in the classroom, or that they lack concrete suggestions of how to change or adapt teaching practice or policy to benefit from their findings.

The approach for how to apply neuroscientific findings to practice forwarded in this paper is to use it primarily to evaluate interventions, rather than individual students (Cortes et al. 2022). If specific brain measures are indeed better indicators of future performance than current behavior, this approach could be used to test the effects of several interventions on a particular brain network and determine which would be more efficient to implement. The major benefit of this approach is that it avoids neuroimaging every child, which is not realistically achievable with current means. Taking this approach could also allow us to refine curricula to optimize any synergistic transfer effects between tasks.

4.9 Conclusions

The primary findings of this study are that although WM is needed for mathematical operations, training VSWM or number-line tasks exclusively, or a mix of both, produces behaviorally relevant differences in effective connectivity that are not evident from analysis of activation levels using the standard GLM approach. Further, these differences arise during training, can be used to differentiate the training conditions, and can predict the rate of improvement on the mathematics tasks during the training period. This means that we can detect the neural changes resulting from different but relatively similar training paradigms, and correlate them with their resulting performance increases, meaning that we can potentially evaluate the effectiveness of interventions by measuring specific connectivity parameters that we want to promote. This could be of value whether we are aiming to improve shared underlying cognitive machinery or specific tasks. Future studies could explore whether these connectivity changes can also be effectively measured using resting-state data in place of task-based fMRI. Since resting state measurements tend to be more consistent than task fMRI, this could be a candidate for individual or curricular evaluation in the future if it can be shown to derive similar enough results to what our task driven DCM has shown.

In conclusion, connectivity measures can be more sensitive than mean brain activity levels, and DCM could potentially be used to evaluate the effectiveness of different classroom interventions on behaviourally relevant connections in the brain.

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Data availability

Data will be made available on request.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to suggest improvements regarding readability of long or complex sentences. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Appendix A

B parameters in stepwise LOO models.

Format: 'B(To,From,condition(2=change over time))', 1=calc, 2=ISPL, 3=rSPL, 4=ISFG, 5=rSFG.

	DCM parameters in model predicting math improvement rate	T (23)	r	p	sig
Original model	B(4,4,2)', 'B(5,5,2)', 'B(3,5,2)', 'B(5,3,2)', 'B(1,1,2)', 'B(2,4,2)'	1.54	0.31	0.068	
gen 2	'B(4,4,2)', 'B(5,5,2)', 'B(3,5,2)', 'B(5,3,2)', 'B(1,1,2)'	1.50	0.30	0.074	
	'B(4,4,2)', 'B(5,5,2)', 'B(3,5,2)', 'B(5,3,2)', 'B(2,4,2)'	-0.59	-0.12	0.721	
	'B(4,4,2)', 'B(5,5,2)', 'B(3,5,2)', 'B(1,1,2)', 'B(2,4,2)'	1.78	0.35	0.044	*
	'B(4,4,2)', 'B(5,5,2)', 'B(5,3,2)', 'B(1,1,2)', 'B(2,4,2)'	1.72	0.34	0.049	*
	'B(4,4,2)', 'B(3,5,2)', 'B(5,3,2)', 'B(1,1,2)', 'B(2,4,2)'	1.57	0.31	0.065	
	'B(5,5,2)', 'B(3,5,2)', 'B(5,3,2)', 'B(1,1,2)', 'B(2,4,2)'	1.58	0.31	0.064	
gen 3	'B(4,4,2)', 'B(5,5,2)', 'B(3,5,2)', 'B(1,1,2)'	1.52	0.30	0.071	
	'B(4,4,2)', 'B(5,5,2)', 'B(3,5,2)', 'B(2,4,2)'	-1.30	-0.26	0.896	
	'B(4,4,2)', 'B(5,5,2)', 'B(1,1,2)', 'B(2,4,2)'	1.83	0.36	0.040	*
	'B(4,4,2)', 'B(3,5,2)', 'B(1,1,2)', 'B(2,4,2)'	1.78	0.35	0.044	*
	'B(5,5,2)', 'B(3,5,2)', 'B(1,1,2)', 'B(2,4,2)'	1.82	0.35	0.041	*
gen 4	'B(4,4,2)', 'B(5,5,2)', 'B(1,1,2)'	1.75	0.34	0.046	*
	'B(4,4,2)', 'B(5,5,2)', 'B(2,4,2)'	-0.98	-0.20	0.831	
	'B(4,4,2)', 'B(1,1,2)', 'B(2,4,2)'	1.94	0.37	0.033	*
	'B(5,5,2)', 'B(1,1,2)', 'B(2,4,2)'	1.83	0.36	0.040	*
gen 5	'B(4,4,2)', 'B(1,1,2)'	1.76	0.34	0.046	*
	'B(4,4,2)', 'B(2,4,2)'	-0.97	-0.20	0.829	
	'B(1,1,2), B(2,4,2)'	1.97	0.38	0.031	*
'gen 6'	'B(2,4,2)'	-1.10	-0.22	0.850	
	'B(1,1,2)'	1.70	0.33	0.051	

Appendix B

Full SPM results table for main effect analysis, thresholded at $p < 0.001$ uncorrected.

cluster			peak				(mm)		
p(FWE-corr)	k	p(unc)	p(FWE-corr)	T	Z	p(unc)	x	y	z
0	1989	2.64E-20	0.00017	7.317108	6.072589	6.29E-10	-8	-90	-4
			0.011767	6.116044	5.307658	5.55E-08	-8	-90	-14
			0.043766	5.698606	5.020098	2.58E-07	-14	-82	-16
5.49E-07	499	1.63E-08	0.03567	5.76483	5.066495	2.03E-07	28	6	64
			0.601161	4.694846	4.280016	9.34E-06	24	2	52
0	2356	1.21E-22	0.055458	5.621181	4.965477	3.43E-07	20	-72	58
			0.071679	5.536151	4.905023	4.67E-07	14	-60	60
			0.689983	4.609299	4.2137	1.26E-05	6	-62	68
1.55E-11	995	4.6E-13	0.056644	5.614215	4.960543	3.51E-07	8	28	40
			0.392078	4.907289	4.442484	4.45E-06	10	28	32
			0.603909	4.692215	4.277983	9.43E-06	-2	18	52
9.34E-07	477	2.77E-08	0.059239	5.599443	4.950068	3.71E-07	32	22	10
			0.220805	5.140244	4.617018	1.95E-06	34	20	2
1.17E-07	565	3.47E-09	0.199521	5.178256	4.645139	1.7E-06	-14	-60	-48

			0.991379	4.090412	3.800491	7.22E-05	-14	-50	-50
			0.991992	4.084432	3.795619	7.36E-05	4	-54	-30
4.84E-09	709	1.44E-10	0.240578	5.107519	4.592727	2.19E-06	-18	-70	62
			0.537534	4.756238	4.32729	7.55E-06	-22	-78	58
			0.988457	4.114958	3.820461	6.66E-05	-22	-58	48
0.162276	79	0.005259	0.344782	4.962868	4.484467	3.65E-06	-28	-68	-32
2.75E-08	629	8.18E-10	0.456425	4.837698	4.389611	5.68E-06	54	12	30
			0.899056	4.36611	4.022383	2.88E-05	54	32	30
			0.980549	4.162793	3.859258	5.69E-05	48	28	38
0.000151	284	4.48E-06	0.623487	4.673474	4.263496	1.01E-05	-26	-4	50
			0.75547	4.543428	4.162288	1.58E-05	-24	0	58
			0.985103	4.137656	3.838891	6.18E-05	-30	-2	64
0.001005	221	2.99E-05	0.951647	4.262465	3.939587	4.08E-05	-30	24	0
			0.982301	4.153722	3.851913	5.86E-05	-32	18	8
			0.999974	3.770314	3.536258	0.000203	-38	16	-8
0.274077	66	0.009514	0.986855	4.126366	3.829728	6.41E-05	-44	24	30
			0.999778	3.863405	3.613829	0.000151	-54	24	36
0.883296	30	0.063806	0.999889	3.831081	3.586961	0.000167	16	-58	-48

			1	3.496384	3.304598	0.000476	14	-58	-56
0.999945	9	0.291346	0.999927	3.81251	3.571492	0.000177	-30	-46	-38
1	2	0.636066	0.999995	3.706827	3.483017	0.000248	16	-46	-54
1	5	0.434558	0.999998	3.682827	3.46282	0.000267	8	-82	-26
0.999731	11	0.244199	1	3.632456	3.420303	0.000313	-36	-66	-46
0.997387	15	0.176648	1	3.612432	3.403354	0.000333	-22	-78	32
0.95682	24	0.093338	1	3.594595	3.388233	0.000352	42	-80	-12
1	3	0.552732	1	3.570374	3.367667	0.000379	-24	-66	-48
1	1	0.751994	1	3.467209	3.27963	0.00052	30	62	-8
1	1	0.751994	1	3.466262	3.278818	0.000521	-18	-36	-40
1	1	0.751994	1	3.463124	3.276129	0.000526	36	-48	74
0.999979	8	0.319784	1	3.420506	3.239544	0.000599	16	-80	-46
1	2	0.636066	1	3.38504	3.209008	0.000666	32	36	38
0.999945	9	0.291346	1	3.378898	3.203712	0.000678	-46	-40	42
1	2	0.636066	1	3.364008	3.19086	0.000709	-12	-86	48
1	2	0.636066	1	3.362264	3.189354	0.000713	8	-22	-12
1	3	0.552732	1	3.345252	3.174653	0.00075	-10	34	24
1	3	0.552732	1	3.334718	3.16554	0.000774	-24	-76	-48

1	1	0.751994	1	3.330266	3.161686	0.000784	-56	20	36
1	3	0.552732	1	3.322001	3.15453	0.000804	-30	-48	44
1	1	0.751994	1	3.306303	3.140923	0.000842	20	-46	-46
1	1	0.751994	1	3.306115	3.14076	0.000843	-8	-74	-50
1	1	0.751994	1	3.291033	3.127672	0.000881	24	-50	78
1	1	0.751994	1	3.275742	3.114389	0.000922	-10	-60	72
1	1	0.751994	1	3.267638	3.107343	0.000944	-32	-80	20
1	1	0.751994	1	3.264472	3.104589	0.000953	-32	-52	52
1	1	0.751994	1	3.255577	3.096849	0.000978	20	-50	-46
1	1	0.751994	1	3.250121	3.092098	0.000994	-34	-82	22